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The Dao, the Hack, the Soft Fork and the Hard Fork

9 comments | Antonio Madeira | 12 Mar 2019 | 75.27 k

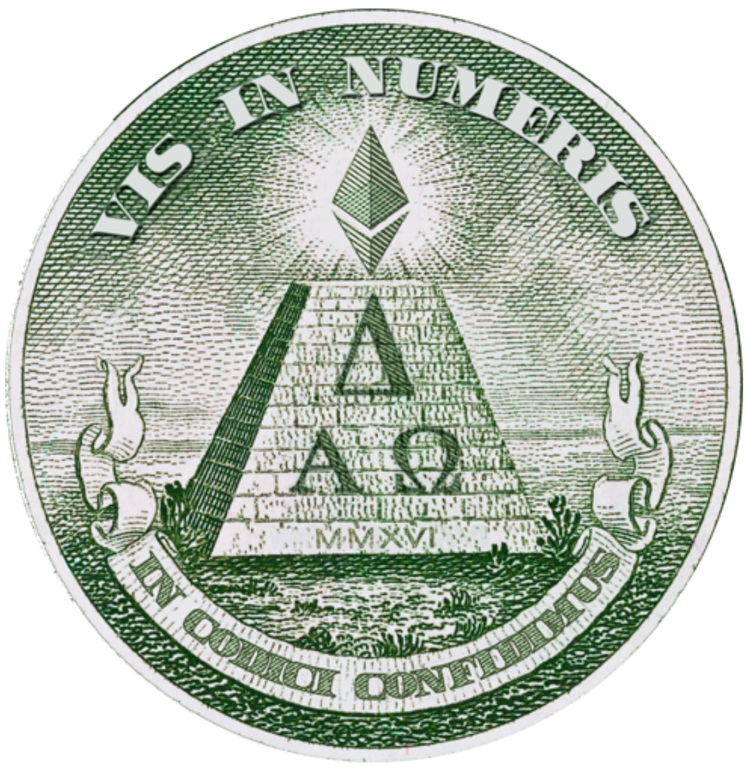
What was the DAO incident?

In this guide we attempt to cover: how and why The DAO was created, how The DAO was exploited, how the soft fork failed miserably, and why everyone was relieved it did so, and how the hard fork led to a split community and the creation of Ethereum Classic.

Part 1: The DAO - Venture Fund Evolution

The Decentralized Autonomous Organization (known as The DAO) was meant to operate like a venture capital fund for the crypto and blockchain space. The lack of a centralized authority reduced costs and in theory provided more control and access to the investors.

At the beginning of May 2016, a few members of the Ethereum community announced the inception of The DAO, which was also known as Genesis DAO. It was built as a smart contract on the Ethereum blockchain. The coding framework was developed open source by the Slock.it team but it was deployed under "The DAO" name by members of the Ethereum community. The DAO had a creation period during which anyone was allowed to send Ether to a special wallet address in exchange for DAO tokens on a 1-100 scale. The creation period was an unforeseen success as it managed to gather 12.7 Ether (worth around $150M at the time), making it the biggest crowdfund ever. At some point, when Ether was trading at $20, the total Ether from The DAO was worth over $250 million.



The DAO was a complex Smart Contract with many features and should have allowed companies to make proposals for funding. Once a proposal was whitelisted by one of the curators, the DAO token holders (aka DAO investors) would then need to vote on the proposal. If the proposal received a 20% quorum - the requested funds would be released into the whitelisted contractor's wallet address. The team of curators that could whitelist addresses was put in place in order to avoid spam proposals and so as to have some human oversight in the automated process. Most of the curators were notable members of the Ethereum community.

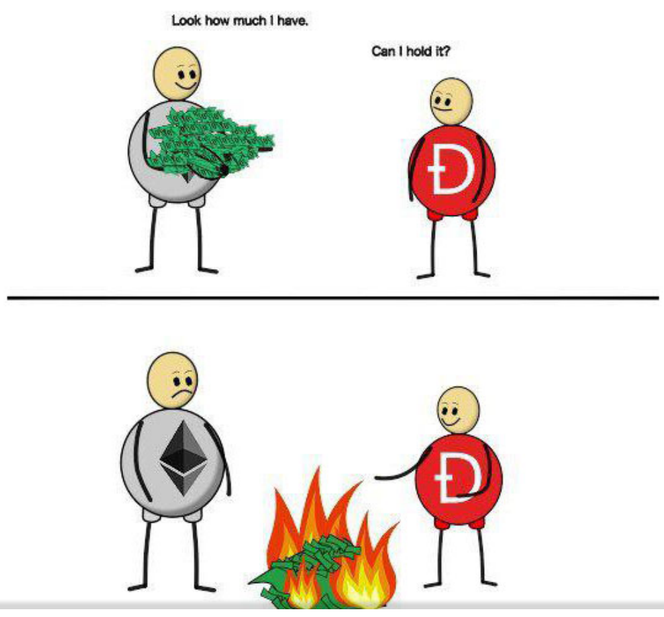
In order to allow investors to leave the organization, in case a proposal that they saw as damaging or of poor quality was accepted, The DAO was created with an "exit door" known as the "split function". This function allowed users to reverse the process and to get back the Ether they sent to the DAO. If somebody decided to split from The DAO, they would create their own "Child DAOs" and approve their proposal to send Ether to an address after a period of 28 days. You could also split with multiple DAO token holders and start accepting proposals to the new "Child DAO".

The DAO launch went smoothly and proposals were created and voted on, security issues were raised during the coming weeks, there was a big community call for a moratorium, but it was not implemented and most of the security issues we not addressed fast enough.

Part 2: The Hack

On the 18th of June, members of the Ethereum community noticed that funds were being drained from The DAO and the overall ETH balance of the smart contract was going down. A total of 3.6m Ether (worth around $70M at the time) was drained by the hacker in the first few hours. The attack was possible because of an exploit found in the splitting function. The attacker/s withdrew Ether from The DAO smart contract multiple times using the same DAO Tokens. This was possible due to what is known as a recursive call exploit.

In this exploit, the attacker was able to "ask" the smart contract (DAO) to give the Ether back multiple times before the smart contract could update its own balance. There were two main faults that made this possible: the fact that when the DAO smart contract was created the coders did not take into account the possibility of a recursive call, and the fact that the smart contract first sent the ETH funds and then updated the internal token balance.



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is not working, it doesn't mean that the internet is not working, it simply means

The hacker stopped draining The DAO for unknown reasons, even though they could have continued to do so.

The Ethereum community and team quickly took control of the situation and presented multiple proposals to deal with the exploit.

Part 3: The Soft Fork - Good news for everyone!

In order to prevent the hacker from cashing in the Ether from his child DAO after the standard 28 days, a soft-fork was voted on and came very close to being introduced. A few hours before it was set to be released, a few members of the community found a bug with the implementation that opened a denial-of-service attack vector. This soft fork was designed to blacklist all the transactions made from The DAO.

Part 4- The Hard Fork - Community split

A more conclusive solution was then put up for vote: the hard fork. This hard fork had the sole function of returning all the Ether taken from the DAO to a refund smart contract. The new contract would have only one function: withdraw. The DAO token holders could request to be sent 1 ETH for every 100 DAO. The investors that had paid more than 1 ETH for 100 DAO could request the difference from the original address. This proposal generated a lot of controversy among the Ethereum community, which was split into 2 groups: hard fork supporters and non-supporters.

The **anti hard-fork** group had the following arguments:

- "Code is law" - the original statement of The DAO terms and conditions should stand under any circumstances
- Things that happen on the blockchain are immutable and they should never change regardless of what the outcome is
- There is a slippery slope and once you modify/censor for one reason there is not a lot to keep you from doing it for other contracts
- The decision to return the money is short sighted and you might reduce the value of ETH down the line based on your decision to act now
- This is a bailout

Users that **supported the hard fork** argued that:

- "Code is law" is too drastic of a statement at the current time and humans should have the final say through social consensus
- The hacker could not be allowed to profit from the exploit as it is ethically wrong and the community should intervene
- The slippery slope argument is not valid as the community is not beholden to past decisions, people can act rationally and fairly in each situation
- It would be problematic to leave such a big piece of the Ether supply in the hands of a malicious actor and it might harm the value of Ether down the line
- This is not a bailout as money isn't being taken from the community, it is just a return of funds to the original investors
- It would stop an ongoing war between the white-hat hackers and the hacker that would demoralize the community and possible continue for many years
- The exploit was big enough to take action and reverse it
- If the community acts now it will make people that are unethical think twice before using Ethereum as their platform of choice
- A hard-fork to return the funds would keep regulators and the legal system out of the debate: "our mess, we fixed it."

In order to reach a quick consensus, the hard fork proposal was voted on and approved by Ether holders, who had to send a transaction to a voting platform. A super majority of people (89%) voted for the Hard-Fork and it took place during the 1920000th block (20th July 2016).

This was when Ethereum Classic was born.

Comments

What is your opinion on The Dao, the Hack, the Soft Fork and the Hard Fork? (You will be banned if you post affiliate / referral links. Please keep the conversation civilized and remember, we're all in this together!)

Post

ron.wild104
3 years ago

Digital Currency is an expert bitcoin miner willing to help those who want to make profit on bitcoin or those who lost their earnings in one way or another trading bitcoin, forex stock, commodities, hackers, and binary options. I give you my word that he will meet up with the following criteria- You can create your own account with his recommended broker, then you have access to it at anytime, fund it and send him the details for management.- 95% guaranteed profit daily- With a minimum investment of $1200 your ROI is estimated at 0.956 btc at the end of 3 trading days. $1900 earns you 2.25 btc and $3950 earns 4.5btc.- Payout within 72 hours after the end of every trading period.he has worked with a lot of people who had lost their btc or Miners who want to make profit fast on btc, he does that for a living. just send him a message on digitalcurrency14 @gmailcom and thank me later

reply Agree

jeremypalafox1
3 years ago

Have you gotten your bitcoins stolen from your wallet or invested in an ICO that turned out to be a scam, you are not alone because this happened to me too. I initially lost $404000 in just three months from Cryptoallday and a couple of others. I contacted the authorities and they referred me to Scams Rescue who helped me recover all my funds within 2 days. I'm speaking up to improve awareness of these cryptocurrency thieves and help as much as i can to reduce victims to the bearest minimum. If you have been a victim, Simply visit www scamsrescue com

reply Agree

oshakezoolah
5 years ago

Almost all users were against the hard fork or didn't vote bc the official vote lasted about 12 hours before the update was pushed with eth foundation's self-bailout set to default setting to catch people off guard.

anti-bailout group argued:
* ethereum foundation developers centrally premined 70% of all coins
* ethereum foundation developers were investors in the DAO and were basically bailing themselves out but most refused to say if they were involved and had to be discovered.
* they were using those coins to damage value of the other chain and not letting market decide independently, making it incredibly unprofitable to dissent from them
* they held centralized funding raised from ICO hostage and refused to use it to fund the chain without the bailout of their money, meaning dissent chain had no future and no funding no matter what
* they set changes as default in client after giving from notice about newly official vote to update only 12 hours. 1/3 of the vote to bail out came from a single account. only ~4% total voted for it. EF internally decided that was enough to use their trust as developers to change settings. Auto-updating node operators wouldn't even notice as they switched to bailout chain.
* they severely punished anyone opt-ing out using methods above making it impossible to do.
* they lied to exchanges to keep the name meaning the other chain had no name or market or market value for days
* dissent chain was at near 1/2 of hash power and 1/2 of tx counts when EF decided to refuse its support at all costs cutting off any last minute options to disagree with them
* ethereum is highly centralized, unethical, and lead by some of the worst people imaginable in crypto, and proof that 70% premine guarantees centralization and ethereum can never be decentralized and have never been decentralized, not a single dapp or decentralized tx is possible on ethereum, forever known as "chain of liars and thieves"
check out /r/ethereumfraud for hundreds of sources backing all this up.

1 reply Agree 3

derkvdbent
5 years ago

Everything breaks down. Crypto currency is in its infancy and will need to evolve to survive. Millennium have heralded in

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able to use crypto to build a fairer global community where everyone can ben fit?

reply Agree 1

webmaster.grumpy
5 years ago

As i predicted in my statements then is that a fork would be the start of the downfall of the crypto currency. They smashed the foundation of crypto currency down, the unbreakable block chain. Solution, just fork it :(

reply Agree 2 1

nucleare
6 years ago

It would seem true to believe that history has its own way of repeating itself...I'm no history major or politician, but it sounds similar to the split between Democrats and Republicans in a way, wouldn't some say?

reply Agree 2 2

soycarozo
6 years ago

who knows ?
maybe is bs or not , what time is it ?

reply Agree 1

cybershaman
6 years ago

it is two-faced when some of the eth-founders make rules (code is law) and after that break them personally (hard fork) just because put their millions in this poor coded "smart" contract and do not want to admit their fault, staying passive despite many warnings and alerts in forums where honest people did exactly pointing to this bug - the above mentioned did do nothing knowing this.

After some user made benefits using the law (the code) they decided to change the law (the code) thus returning their own millions and ruining the confidence and trust in the whole concept - the distributed autonomous thing, the ethereum and his blockchain, and all the sponsors and contributors.

After this history-rewriting they are not more and not less than all other fiat-money institutions. "when somebody puts his money in the bank/eth/dao, he is no more owner of them, the bank/eth/dao owns them". In my country there is a sentence/moral: he is not crazy who eats the cake, but he who gives him.

Somebody who think of his own few millions acted against the whole crypto-comunity, ruining the trust. What is the value, the power, and the future of some cryptocoin or fiat coin without trust? Bending laws is always destructive, and humans, survived after the slavery of communism know perfectly that, opposite to naive "protesters" living in political and economical stability.

That's why i am definitely against the law-forking, it is exactly like in Orwell's book "1984" and has only one name - history rewriting. I remain the same bitcoin-believer, and do not trust in such populist actions, like ethereums constitution replacement, made in favor of some personally interested, who are in power.

In my times i destroyed the future possibility for cultural and economical isolation of my people in my country, known as "iron curtain", founding the regional parts of fidonet and the fourth public access to internet in Europe. Now my brothers and sisters breed internet, and cannot be deceived so easily like in isolation times of dictatorship. Never again would this happen in the freest part of cyberspace - the cryptocoin world - it depends of anyone of us, and mostly depends from the personal need for freedom.

View Edits

4 replies Agree 5 1

monty.kirkman
6 years ago

If the thief had of stopped at 5-10 mil I doubt there would have been a HF

2 replies Agree 1

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